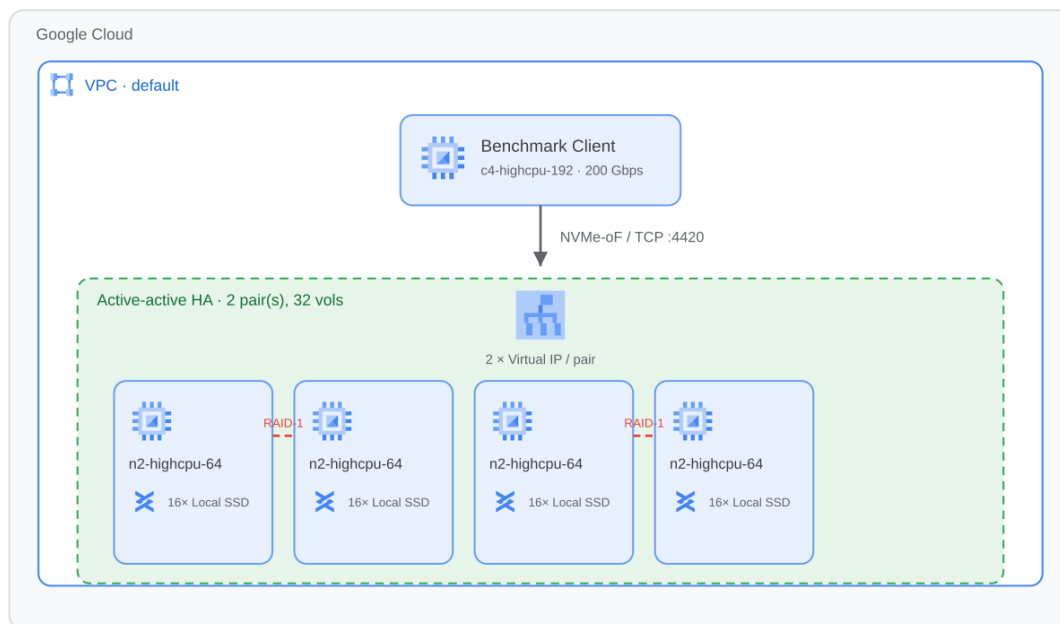


# MayaScale Block — MLPerf Storage v3.0 (UNet3D)

**Submitter:** ZettaLane Systems · **Division:** Closed · **System:** mayascale-gcp\_c4hc192\_n2hc64\_1c\_4s

## architecture

MayaScale — NVMe-oF Block on Google Cloud



### MayaScale Storage Cluster (4 nodes, 2 active-active HA pairs)

- 4 × GCP n2-highcpu-64 (Intel Xeon, 64 vCPU / 2×16 cores, 62 GB RAM)
- Rocky Linux 9.7 · 75 Gbps networking · NVMe-oF/TCP target
- 16 × Local SSD (375 GiB, NVMe TLC) per node (64 total, 23.44 TiB raw → 11.72 TiB usable after mirror)
- Cross-node RAID-1 mirror (nvmetmirror) → 32 mirrored NVMe-oF volumes exported

### Benchmark Client

- 1 × GCP c4-highcpu-192 (Intel Xeon Platinum 8581C (Emerald Rapids), 192 vCPU, 378 GB RAM)
- Ubuntu 24.04 · 200 Gbps networking

- NVMe-oF/TCP initiator → mdadm RAID0 over mirrored volumes (RAID10) → XFS

## Results

- UNet3D (na=4, 4× B200): Accelerator Utilization 93.31%

## Availability

MayaNAS and MayaScale can be deployed from the Google Cloud Marketplace portal or as Terraform modules (<https://github.com/zettalane-systems/open-lustre-cloud>). Contact: [info@zettalane.com](mailto:info@zettalane.com).